

How Transistors Work

What cells are to your body, the transistor is to the chip

Team Digit

If we were to identify one component as “the building block of computers,” it would have to be the Transistor. Invented at Bell Labs in 1947, the transistor is a small electronic device used in all things digital—from your digital watch to supercomputers. We therefore think some understanding of how a transistor operates is required.

To begin, one needs to know a little about semiconductors. These fall under a class of materials that behave as either conductors or insulators depending on certain conditions. Silicon, Germanium, and Gallium Arsenide are the most commonly-used semiconductors. A semiconductor can be of two types, an N-type—which has “excess” electrons—and a P-type material, which has a deficiency of electrons. The N or P is determined by an “impurity” element that is added to a pure semiconductor. If the impurity has more free electrons than the pure semiconductor, the resultant material is a N-type, and if it has less free electrons, you get a P-type material.

The simplest of semiconductor devices is a diode, which is formed when a P-type and an N-type material are brought together, adjacent to each other. The

electrons from the N-type fill up the deficiencies (called “holes”) in the P-type, which leaves a neutral layer at the junction that does not conduct. This “barrier” is overcome by applying a small positive voltage to the diode. When the voltage is removed, the barrier is recreated, making the device not capable of conducting. Similarly, when a negative voltage is applied, the device does not conduct.

The transistor, in a very simplistic way, can be thought of as two diodes joined back to back. We can therefore have two kinds of transistors—N-P-N and P-N-P. Both these configurations can be used in devices; NPN is usually chosen for explanation.

The three blocks that make up a transistor are the Emitter, the Base, and the Collector. The Emitter and Collector are made from the same type of semiconductor, but you can’t interchange their connections—their width and the way they are manufactured are different.

When the blocks come together, two barriers are formed—between the Emitter and Base, and between the Base and Collector. Typical transistor operation sees the Emitter–Base junction “forward-biased”—this means the positive of the battery is connected to the Base, which is P-type, and the negative battery terminal is connected to the Emitter, which is N-type. This ensures that the barrier between these two terminals is eliminated. Similarly, the Base–Collector junction is “reverse-biased.”

Now, electrons are pushed from the Emitter into the Base. The Collector has a positive voltage applied to it, which therefore draws (the negatively-charged) electrons towards it. Thus the electrons that come to the Base from the Emitter are

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drawn into the Collector, which leads to a current flow in the device. The device is now in the On state. When the voltage applied to the Emitter–Base junction is removed or its polarity reversed, the barrier at the Emitter–Base junction reappears, and the current flow stops. The transistor is now in the Off state.

This switching action is the most important capability of a transistor, which is used in digital circuits such as those in computers. Computers work by treating all data as a series of 1s and 0s, or On and Off states. Transistors are used to perform complex calculations based on long series of 0s and 1s: a 32-bit CPU can add, subtract and perform several functions on data that contains a total of 32 0s and 1s.

For each 0 or 1 that needs to be stored or operated upon in a CPU, one transistor is needed. In order to store two 64-bit numbers, a total of 128 transistors is required. This is only to store the numbers during calculation: for each operation such as multiplication or comparison, a thousand or more transistors are required. Today’s CPUs have millions of transistors to perform the myriad calculations that make games, video editing, and so on work for us. All these millions of transistors are basically working in a switching mode—in an On or an Off state at any given time.

Entire books have been written on transistors, and we’ve just taken a cursory look. Over the decades, continued innovation has happened, with the result being smaller sizes, lower power consumption, higher speeds, and reliability. As we enter the age of nanotechnology, we’re likely to see even smaller and newer kinds of transistors—which create the 1s and 0s we’re so dependant on ■

readersletters@jasubhai.com

